



Antimicrobial resistance genes and associated mobile genetic elements in *Escherichia coli* from human, animal and environment

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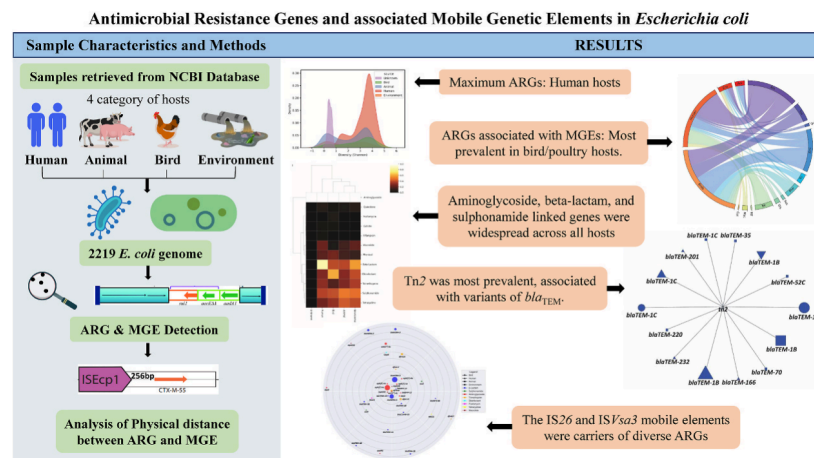
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HIGHLIGHTS

- Human hosts harboured the highest number of ARGs, followed by bird hosts.
- ARGs such as *aph(6)*-Id, *aph(3'')*-Ib, *bla*_{CTX-M}, *bla*_{NDM} were widespread across all hosts.
- The IS26 and ISVsa3 MGEs were carriers of diverse ARGs.
- Tn2 was the most prevalent, majorly carried by IncFIB plasmids.
- ARGs like *bla*_{NDM}, *bla*_{CTX-M} variants displayed strong association with IS elements.

GRAPHICAL ABSTRACT



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ABSTRACT

The global rise of antimicrobial resistance (AMR) poses a significant threat to human health. The environment plays an essential role in transmission of antimicrobial resistance genes (ARGs) between human and animal. Bacterial communities harbour diverse ARGs, carried by mobile genetic elements (MGEs) like plasmids and

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insertion sequences (ISs). Here, a total of 2199 *Escherichia coli* (*E. coli*) whole genome sequences from human, animal, bird and environment were retrieved globally to investigate ARG prevalence and assess their genetic framework. The study highlights how the genetic background including plasmids, IS elements and transposons surrounding ARGs influences their transmission potential. The maximum number of ARGs was found in United Kingdom followed by USA, majorly in human hosts. However, IS-associated ARGs were most prevalent in bird hosts. ARGs like *aph(6)-IId*, *aph(3')-Ib*, *bla_{CTX-M}*, *bla_{NDM}* were widespread across all hosts. Tn2 was the most prevalent, majorly carried by IncFIB plasmids. The IS26 and ISVsa3 carried diverse ARGs, primarily linked to aminoglycoside and β -lactam resistance. The combinations like *mph(A)_{IS6100}* and *bla_{NDM-5}_{IS5}* showed fixed IS-ARG associations. ARGs like *bla_{NDM}*, *bla_{CTX-M}* variants displayed strong association with IS elements. The study highlights possible mechanism of transmission due to close proximity of AMR genes to MGEs, offering promising strategies to combat AMR by predicting and addressing future resistance determinants.

1. Introduction

Escherichia coli (*E. coli*) is a common commensal organism but also exhibit adaptability as a versatile pathogen (Bonten et al., 2020). *E. coli* is widely distributed in the natural environment, including food-producing animals, farm waste materials, untreated water waste, aquaculture, river, soil, human faeces, animal faeces and other related areas (Bonten et al., 2020; Poirel et al., 2018). *E. coli* serves as a reservoir for the acquisition and dissemination of antimicrobial resistance genes (ARGs) and thus, poses a significant challenge globally, impacting both human and animal health (Poirel et al., 2018).

Human activities, especially in agriculture and healthcare, are major reasons causing the ubiquitous pollution of ARGs in the environment (Andersson et al., 2020). In agriculture, antibiotics used in livestock can lead to the emergence of resistant bacteria, which may contaminate food sources and transfer resistance to humans through food or direct contact (Acar and Moulin, 2013; Xu et al., 2022a,b). In healthcare, overuse and misuse of antibiotics in regions with unregulated antibiotic sales, results in increased resistance in humans (da Silva et al., 2021). In hospitals, patients treated with antibiotics may develop resistant infections and spread them to others, perpetuating the antimicrobial resistance (AMR) cycle. (Ho et al., 2024; Otaigbe and Elikwu, 2023). The interconnected transmission of AMR across farms, healthcare facilities, wastewater plants, and the natural environment creates a One Health challenge (Emara et al., 2023). One health approach recognizes that human, animal, and environmental health are interdependent (Ferri et al., 2017). Key strategies include coordinated monitoring of antibiotic use, food chain risk assessment, development of animal-specific antibiotics, promoting stewardship programs, investing in research and development for new antimicrobial options (Ahmed et al., 2024; Hernando-Amado et al., 2019).

Antibiotics, vital in modern medicine, were discovered in the 1930s–1960s (Durand et al., 2019). However, the era ended with the emergence of resistant bacteria, including *E. coli* (Founou et al., 2022). β -lactam are the most widely used antibiotics for treating *E. coli* infections in both humans and animals due to their broad-spectrum activity, low toxicity, and high tolerance (da Cunha et al., 2019). However, Gram-negative bacteria can develop resistance to β -lactams by producing β -lactamase enzymes (Rana et al., 2022). There are certain ARGs, including *bla_{CTX-M}* and *bla_{NDM}* gene variants, that have emerged clinically and have been disseminated globally via environment, human and animal populations (Acman et al., 2022; Castanheira et al., 2011; Kumarasamy et al., 2010; Rana et al., 2022; Yu et al., 2024). This results in decreased effectiveness of β -lactam antibiotics in treating *E. coli* infections, posing a significant public health threat (Castanheira et al., 2011; Rana et al., 2022; Tian et al., 2020). As a result, standard treatments become ineffective, leading to higher rates of treatment failure, prolonged illness, higher healthcare costs due to extended hospital stays and increased morbidity (Ahmed et al., 2024). This emphasizes the urgent need to tackle antibiotic resistance.

Bacterial communities in human, animal and environment harbour various ARGs (Inda-Díaz et al., 2023). Mobile genetic elements (MGEs) like plasmids, insertion sequences (ISs) and transposons (Tns) enable

E. coli to acquire ARGs from other bacteria through Horizontal gene transfer (HGT) (Lipszyc et al., 2022; Zhao et al., 2020). Plasmids, carrying multiple ARGs, can transfer between bacteria via conjugation, starting at the origin of transfer (*oriT*), allowing bacterial populations to acquire resistance (Castaneda-Barba et al., 2023). In *E. coli*, the ARG carrying plasmids can integrate into the host chromosome via homologous recombination at ISs, creating high-frequency recombination (*Hfr*) strains (Vos et al., 2024). Transposons (Tns) and ISs are “jumping genes” that can autonomously move within the genome and promote the mobilization of ARGs (Partridge et al., 2018). They primarily use a non-replicative mechanism, aiding in the spread of ARGs across different species (Lipszyc et al., 2022; Partridge et al., 2018). ISs, small (~2.5 kb in length) transposable sequences, encode a transposase enzyme and can form composite transposons, with two flanking ISs (Acman et al., 2022). This facilitate insertion and excision of ARGs enabling their introduction to new locations within the genome (Acman et al., 2022; Partridge et al., 2018). The IS elements act as strong promoters for ARG expression (Hoeksema et al., 2018; Partridge et al., 2018). By mobilizing ARGs, MGEs help *E. coli* to adapt to diverse hosts and antibiotic-rich environments, driving resistance spread across human, animal, and environmental interfaces (Partridge et al., 2018).

The presence of an ARG in a genome does not necessarily imply resistance, and the location in which an ARG is present is essential in determining its function (Nielsen et al., 2022). Hence, it is necessary to study the genetic framework around ARGs to assess its mobility pattern and its impact on human health (Partridge et al., 2018). The association between the ARGs and IS elements has been previously acknowledged in various studies (Cao et al., 2022; Evans et al., 2020; Leekitcharoenphon et al., 2021; Nielsen et al., 2022; Razavi et al., 2020). The overall genetic background surrounding all ISs displayed more variability in terms of ARGs. The IS elements, notably IS6, Tn3, IS4, and IS1, were prevalent and exhibited strong associations with diverse ARGs (Razavi et al., 2020). It has been observed that several classes of genes encoding broad-spectrum β -lactamases exhibit high mobility (Nielsen et al., 2022).

Analyzing the proximity between ARGs and MGEs provides insight into transmission dynamics by highlighting the potential for ARGs to be mobilized (Partridge et al., 2018). When closely situated, ARGs and MGEs are more likely to be co-transferred via HGT, enhancing the spread of resistance across bacterial populations (Karkman et al., 2019; Nielsen et al., 2022). MGEs serve as recombination hotspots, so ARGs near MGEs are more readily integrated into new genomes, further promoting their dissemination (Partridge et al., 2018). Therefore, a more targeted effort is needed to comprehend the association between MGEs and ARGs. The present study assessed the prevalence of ARGs in *E. coli* from human, animal, bird, and environment hosts, using publicly available data from NCBI. We analysed the physical distance between ARGs and MGEs to assess the transmission potential. To the best of our knowledge, sharing of ARGs and IS element combinations across different hosts by analysing the physical distance between ARGs and IS elements has not been explored previously. Our study sought to investigate the presence of IS-linked ARGs with consensus distance between them found across different hosts, shedding light on their transmission

potential. Understanding the specific IS elements that tend to associate with ARGs can serve as an initial step in the discovery of novel ARGs through predictive models. Our study offers a way to rank ARGs by their dissemination risk, helping to identify AMR hotspots for focused monitoring/surveillance. This approach supports proactive strategies to limit ARG transmission in high-risk settings like hospitals and agriculture.

2. Methods

2.1. Data acquisition

A total of 2199 *E. coli* whole genome sequencing dataset generated from Illumina sequencing platform, were retrieved from the NCBI on January 2, 2023 (Projects: PRJNA292663, PRJNA473419, PRJNA658789, PRJNA750276, PRJNA788733, PRJNA805266, PRJNA812750, PRJNA871076, PRJNA891712, PRJNA892652). The isolates were retrieved from various sources and countries and were categorized into human, animal, bird and environment host (Supplementary Table S1, S2 & S3). The raw read data was retrieved in the SRA file format using SRA toolkit NCBI.

2.2. Data analysis and ARG calling

To ensure high-quality sequencing data prior to assembly, FastQC and Trimmomatic pipelines were implemented. The FASTQC v0.11.9 (<https://github.com/s-andrews/FastQC>) provided metrics such as read quality, GC content, and adapter contamination. Based on these metrics, low-quality bases at the ends of reads or adapter contamination was identified. The raw reads were adapter-trimmed using Trimmomatic version 0.39 with the following parameters: SLIDINGWINDOW:4:20 MINLEN:25 (Bolger et al., 2014). These parameters removed bases with an average quality below 20 in a 4-base window. Reads shorter than 25 bases were discarded to prevent mis-assemblies, resulting in clean and high-quality dataset ready for assembly (Li et al., 2024a,b; Munk et al., 2022; Onimaru et al., 2018).

The data was de novo assembled using SPAdes 3.15.4 with a Phred score of Q30 (spades.py -careful - 1 forward.fastq - 2 reverse.fastq) (Prjibelski et al., 2020). The assembly quality was assessed with QUAST, focusing on N50 and contig metrics (Mikheenko et al., 2018). The completeness and potential contamination within the genome was assessed using BUSCO v5.4.7 with the bacteria_odb10 database (Manni et al., 2021). Further, the assemblies with genome sizes exceeding 6 Mbp or falling below 4 Mbp were excluded (Bharat et al., 2022; Huang et al., 2024) (Supplementary Table S4).

The ARG-Online Analysis Pipeline (ARGs-OAP v3.2.3) was used to quantify the relative abundance (Reads per Kilobase per Million: RPKM) and diversity of AMR genes against the customised reference RESFINDER database (Release date May 24, 2022) using thresholds (E-value: $1e^{-7}$, identity >90%, hit length ratio 85%) (Yan et al., 2024; Yin et al., 2023). Initially, candidate ARG sequences were filtered using UBLAST, followed by BLASTX alignment of ARG-like sequences against the RESFINDER database (Supplementary Tables S5 and S6) (Yin et al., 2023). The RESFINDER database contained 3204 non-redundant reference resistance genes (Bortolaia et al., 2020) (Supplementary Table S7).

2.3. Detection of contigs carrying ARGs

The datasets were annotated for ARG contigs against the latest version of ResFinder 4.0 using Staramr pipeline (<https://github.com/phac-nml/staramr>) (last updated in May 2022) (Bharat et al., 2022; Bortolaia et al., 2020). To increase the accuracy of ARG hits, the E-value cutoffs were set to $10e^{-5}$, >90% query coverage and >90% (Calderón-Franco et al., 2022). The plasmid contigs were identified using PlasmidFinder database through staramr pipeline (<https://github.com/phac-nml/staramr>) (Supplementary Table S8). PlasmidFinder matches contig sequences to known plasmid replicons included in the

PlasmidFinder database (Founou et al., 2022). Replicons are specific regions (typically <1 kb) responsible for plasmid replication and regulation (Carattoli and Hasman, 2020). The PlasmidFinder database currently contains 133 unique replicon sequences, allowing it to accurately identify plasmid incompatibility group (Bharat et al., 2022; Carattoli and Hasman, 2020). To increase classification confidence, the filtering criteria, >90% sequence identity and query coverage was applied. Any contigs not identified as plasmid-origin by PlasmidFinder were classified as chromosomal.

2.4. Detection of mobile genetic elements

The IS elements and transposons were identified using MobileElementFinder written in Python v3.7. The MobileElementFinder tool facilitates the detection of MGEs within the assembled genome by BLAST search to the reference MGE database (Johansson et al., 2021). The final MGE database comprises approximately 4450 MGE sequences from around 1050 different species, enabling the tool to detect MGEs across a wide range of bacterial species, including *E. coli* (Johansson et al., 2021). The Mobile elements with threshold values of >90% query coverage and >90% similarity was filtered using an in-house python script, to enhance the reliability and accuracy of MGE identification (Cheng et al., 2024; Fan et al., 2013).

2.5. Detection of associated ARGs with MGE

The MGEs and ARGs were considered associated when the distance between the coordinates of MGE and ARGs was <10 kb, both upstream and downstream (Cao et al., 2022; Tóth et al., 2021; Trosvik et al., 2024). Various combinations were made where both the ARGs and IS elements were located on the same contig within 10 kb distance. The close proximity of IS elements and ARGs increases the likelihood of their co-mobilization through HGT, indicating a stable and long-term association (Boochandani et al., 2022; Cao et al., 2022; Evans et al., 2020; Karkman et al., 2020; Partridge et al., 2018; Razavi et al., 2020). Thus, four categories of IS-ARG associations based on distance were made: (i) <1 kb – very strong association (ii) 1 kb–3kb – strong association (iii) 3 kb–5kb – moderate association (iv) 5 kb–10kb – weak association. The association was depicted through radial plots for each IS element found in different hosts using netgraph and matplotlib. The distance was represented in square root transformation with a maximum distance of 10 kb.

2.6. Statistical analyses

The Shannon, Simpson and Chao1 indices were used to assess alpha diversity of ARGs. The β -diversity was calculated based on Bray-Curtis dissimilarity matrix using the raw matrix data and visualized through PCoA (Principle Coordinate Analysis) using SciPy and scikit-bio package in python. Analysis of Similarity (ANOSIM) was used to test the significant differences in microbial community between different hosts with 999 permutations and P value < 0.05 using Scikit-bio and SciPy package in python.

All bar plots, network plots, bubble charts were made in python using matplotlib library. Heatmap was made using PyComplexHeatmap package (Ding et al., 2023).

3. Result

3.1. Sample characteristics

A total of 2199 *E. coli* WGS samples from diverse hosts (human, animal, bird and environment) across different geographical locations were retrieved and analysed for the presence of ARGs and MGEs. The highest number of isolates was found in human samples (1065, 48.43%) followed by animal (511, 23.23%), bird (247, 11.23%) and environment

samples (190, 9.32%) (Supplementary Table S2). A few of them were of unknown host origin (186, 8.45%). The isolates were retrieved from 11 countries, the maximum being from United Kingdom (1212, 55.11%) followed by USA (373, 16.9%) (Supplementary Table S3).

3.2. Abundance and diversity of ARGs

All the ARG types were classified according to the mechanism of action and grouped into their respective antibiotic classes (Supplementary Table S7). The AMR gene abundance varied across different hosts and countries. The number of observed ARGs was maximum in United Kingdom, followed by USA. Resistome diversity indices were calculated for each sample based on Chao1, Shannon and Simpson metrics using the normalized abundance data (RPKM). Resistome abundance was significantly higher in human hosts as compared to other hosts (Analysis of Similarity, ANOSIM test, $P < 0.05$). Overall, all the metrics exhibited a higher median evenness in human host samples (Figs. 1 and 2). The PCoA assessed the differences in resistome revealing the clustering and dissimilarity patterns of ARG compositions among human, animal, bird, and environment contributing to 15.32% microbiome variation (Supplementary Fig. S1).

The *bla*_{TEM}, *aph*(6)-Id, and *aph*(3'')-Ib ARGs were ranked highest in terms of their relative abundance. ARGs encoding resistance to aminoglycoside (majorly *aph*(6)-Id), β -lactam (*bla*_{TEM} and *bla*_{CTX-M} variants) and disinfectant (mainly *qacE*) were abundant in most samples while those to tetracycline, sulphonamide and trimethoprim were found at different levels in different hosts (Fig. 3). There was a relatively higher levels of ARGs conferring resistance to aminoglycoside and β -lactam in UK humans (Supplementary Fig. S2).

3.3. Abundance of contigs carrying ARGs and MGEs

A total of 15,748 contigs carrying ARGs categorized into 210 distinct types of ARGs were detected (Supplementary Tables S9 and S10). All the ARGs were grouped into their respective classes based on hosts and countries origin, resulting in the representation of 12 AMR classes (Fig. 4a and Supplementary Fig. S3).

The human hosts harboured the maximum number of contigs carrying resistance genes followed by animal and bird hosts. A consistent high prevalence of ARGs conferring resistance to aminoglycoside, β -lactam, and sulphonamide were detected across all host types (Fig. 4a). More than 75% of the total ARG abundance (12,450/15,748) was attributed to twenty-seven distinct ARGs. Notable among these were *qacE* (830, 6.66%), *sul1* (822, 6.60%), *sul2* (814, 6.53%) and *aph*(6)-Id (779, 6.25%) (Fig. 4b & Supplementary Table S9). This trend was particularly noticeable in samples obtained from human hosts. The country-wise data showed a consistent high abundance of aminoglycoside, β -lactam, and sulphonamide resistance genes across all countries. UK showed a high ARG prevalence (9863/15461, 63.8%) compared to the USA (1205/15461, 7.8%). Specifically, ARGs corresponding resistance to aminoglycoside such as *aac*(6')-Ib-cr (473/562) and *aph*(3'')-Ib (475/791) and β -lactam like *bla*_{CTX-M-15} (305/390) and *bla*_{NDM-5} (114/

118) were highly prevalent. However, USA showed a lower prevalence and diversity of these ARGs, with *aac*(6')-Ib-cr (56/562), *aph*(3'')-Ib (49/791), *bla*_{CTX-M-15} (55/390), and *bla*_{CTX-M-14} (8/94). This lower ARG prevalence in USA could be attributed in part to the smaller sample size compared to UK (Supplementary Table S3).

A total of 24744 contigs carrying IS elements were found within the *E. coli* genome, of which, *ISec1* (2091, 8.45%), *IS609* (1799, 7.27%), *ISec38* (1311, 5.29%) and *ISKpn8* (1239, 5%) were the most prevalent (Fig. 4c). The *IS26* (1042, 4.2%) and *IS629* (1035, 4.18%) was found maximum in human isolates.

3.4. Association of ARGs with MGEs (IS elements and transposons)

The IS elements facilitate the transfer of genetic elements such as ARGs across different bacterial species. To explore the transferability patterns of resistance genes across different hosts and countries, IS-associated ARGs were identified. Since plasmids play a key role in transmission of ARGs, IS-linked ARGs of plasmid origin were identified.

These IS-associated ARGs were most abundant in bird samples, followed by human samples. A total of 464 out of 15,748 ARGs were found associated with diverse IS elements, i. e., they were present within 10 kb of IS elements. The primary IS-associated ARGs include *floR* (35, 7.52%), *sul2* (34, 7.31%) and *tet*(B) (33, 7.09%) (Supplementary Table S8). However, the clinically important carbapenemases found associated with IS elements include *bla*_{NDM} variants ($n = 43$): *bla*_{NDM-5} ($n = 27$), *bla*_{NDM-1} ($n = 11$), *bla*_{NDM-7} ($n = 4$), *bla*_{NDM-4} ($n = 2$), *bla*_{CTX-M} variants ($n = 31$): *bla*_{CTX-M-55} ($n = 7$), *bla*_{CTX-M-14} ($n = 6$), *bla*_{CTX-M-1} ($n = 5$), *bla*_{CTX-M-15} ($n = 4$), *bla*_{CTX-M-32} ($n = 4$), *bla*_{CTX-M-65} ($n = 3$), *bla*_{CTX-M-2} ($n = 2$) and *bla*_{OXA-10} ($n = 3$). The host and country profile of these ARGs were diverse. The *bla*_{NDM} and *bla*_{CTX-M} gene variants associated with IS elements were predominantly found in human hosts in UK. The IS-associated ARGs conferring resistance to phenicol and trimethoprim were primarily identified in birds.

Specific correspondences between IS-associated ARGs and plasmid Inc types found in different hosts globally were identified (Fig. 5b). Of the IS-associated ARG contigs, 32.68% (152/465) were present on plasmids, while 67.31% (313/465) were of chromosomal origin. (Supplementary Fig. S4). The IS-associated ARGs carried by plasmids were majorly of human origin, conferring resistance to tetracycline and β -lactam majorly. A total of 22 incompatibility types (Inc) plasmids were found to carry IS-associated ARGs, with IncFIB(pB171) (27, 22%), IncX3 (19, 15.44%) and IncC (13, 10.56%) accounting for the top three Inc types. The 85.2% (23/27) of the IncFIB(pB171)-type plasmid carried *tet*(B) genes conferring resistance to tetracycline found in human hosts of UK majorly. Only *bla*_{NDM-5} (17, 89.4%) and *bla*_{NDM-1} (2, 10.5%) was found on IncX3-type plasmid in UK human. The IncC-type plasmid carried diverse ARGs corresponding resistance to β -lactam (*bla*_{NDM-1}, *bla*_{NDM-4}), aminoglycoside (*armA*, *rmtC*) and MLS (*msr*(E), *mph*(E)). The *bla*_{CTX-M-32} isolated from animal source in France and *bla*_{CTX-M-15} isolated from human source in USA was found on IncFIB(AP001918) and IncFIB(pB171)-type plasmids respectively. The *E. coli* isolates found in birds in China, conferring resistance to various ARG classes, were identified on

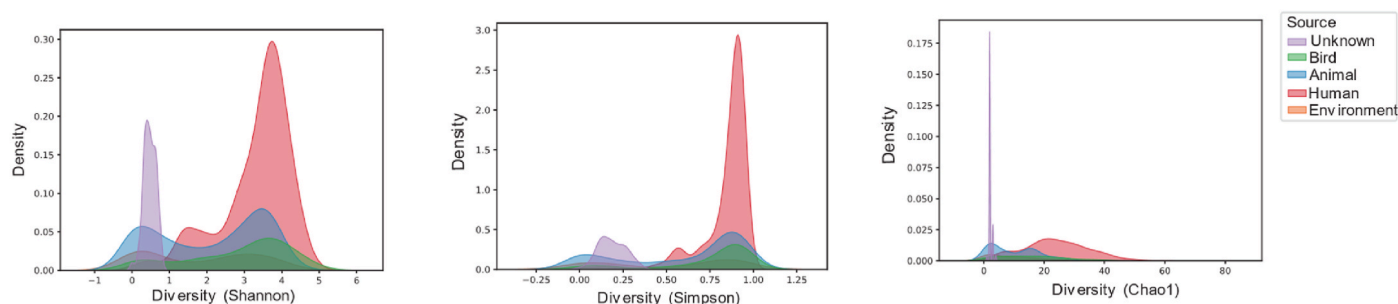


Fig. 1. Kernel density plots for dissimilarity distribution of ARGs across different hosts (a) Simpson diversity (b) Shannon diversity (c) Chao1 diversity.

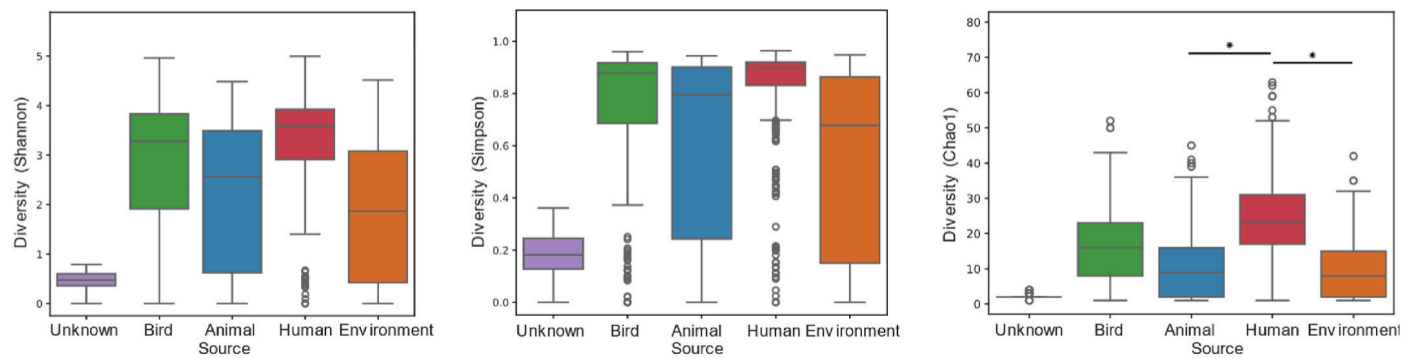


Fig. 2. Box plots for dissimilarity distribution of ARGs across different hosts (a) Simpson diversity (b) Shannon diversity (c) Chao1 diversity. The boxes illustrate the interquartile range (IQR), indicating the spread of the middle 50% of the data, with the upper edge of the box representing the third quartile (75th percentile) and the lower edge representing the first quartile (25th percentile). The horizontal line inside the box marks the median, or the midpoint of the data. The whiskers extend from the edges of the box and depict the range of values within 1.5 times the IQR. The whisker's endpoint marks the lowest and highest values within this range. Any data points beyond this 1.5 times IQR range are denoted by "•". The level of significance was assessed using ANOSIM test with 999 permutations (* $p < 0.05$).



Fig. 3. Heatmap depicting the relative abundance of ARGs clustered by groups of hosts in columns and ARG class in rows. The relative abundance of each ARG has been normalized using the ratio of each ARG RPKM value within all ARGs of the respective column.

diverse plasmid types.

Taxonomic classifications were assigned to contigs containing unit transposons, and their occurrence on plasmid and chromosomal contigs was examined. A total of 2194 contigs carrying unit transposons associated with ARGs were identified, of which 59 transposons were present on plasmids and 2135 were of chromosomal origin. The transposons were classified into 19 types of unit transposons, of which three were found consistently in high abundance in all the hosts namely, Tn2, Tn6024, Tn4656 (Fig. 5c). A total of 5 types of unit transposons (Tn2, Tn801/TnA, Tn1721, Tn7 and Tn4401) were carried on 20 different types of plasmids, carrying diverse ARGs (Fig. 5d). The Tn2 was the most prevalent and carried by 15 different plasmids, predominantly by IncFIB (AP001918) type plasmid. Tn2 was found with maximum size of around 5 kb, majorly associated with variants of *bla*_{TEM}. Another 14 kb Tn7 was found harbouring *dfrA1* and *aadA2* genes (Fig. 5e and f).

3.5. Physical distance between ARGs and IS elements in *E. coli* genome

Examining the genetic environment surrounding the ARGs is a hallmark for investigating mobility pattern of genetic elements among diverse hosts. To investigate the genetic distance between ARGs and IS elements, 464 combinations were obtained. The IS elements displayed

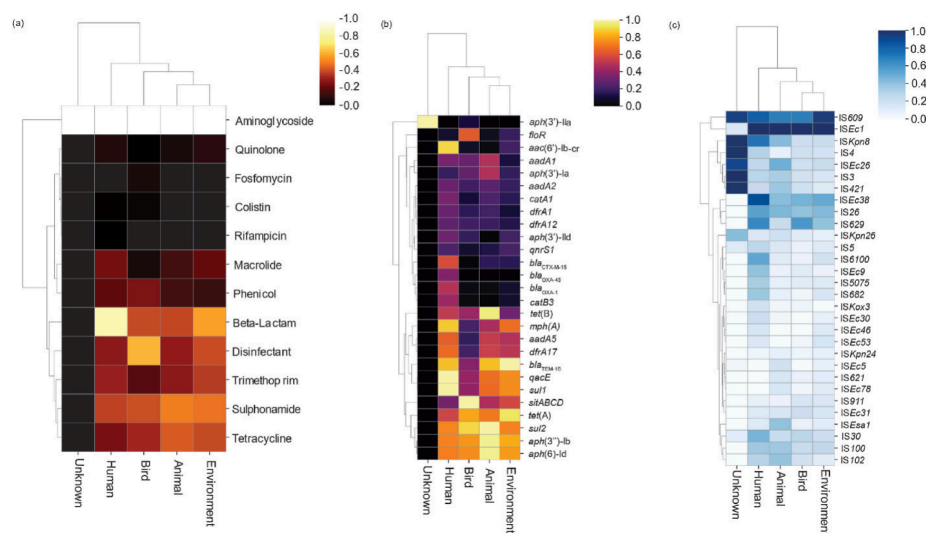


Fig. 4. Heatmaps representing abundance of ARG and IS element carrying contigs. Two-way clustering was applied to both the columns and rows for organizing and visualizing the data. (a) Heatmap depicting frequency of contigs containing ARGs clustered into the respective AMR class grouped by hosts in columns and AMR classes in rows. (b) Heatmap depicting frequency of twenty-seven most abundant contigs containing ARGs grouped into hosts. (c) Heatmap depicting frequency of contigs containing IS elements grouped by hosts.

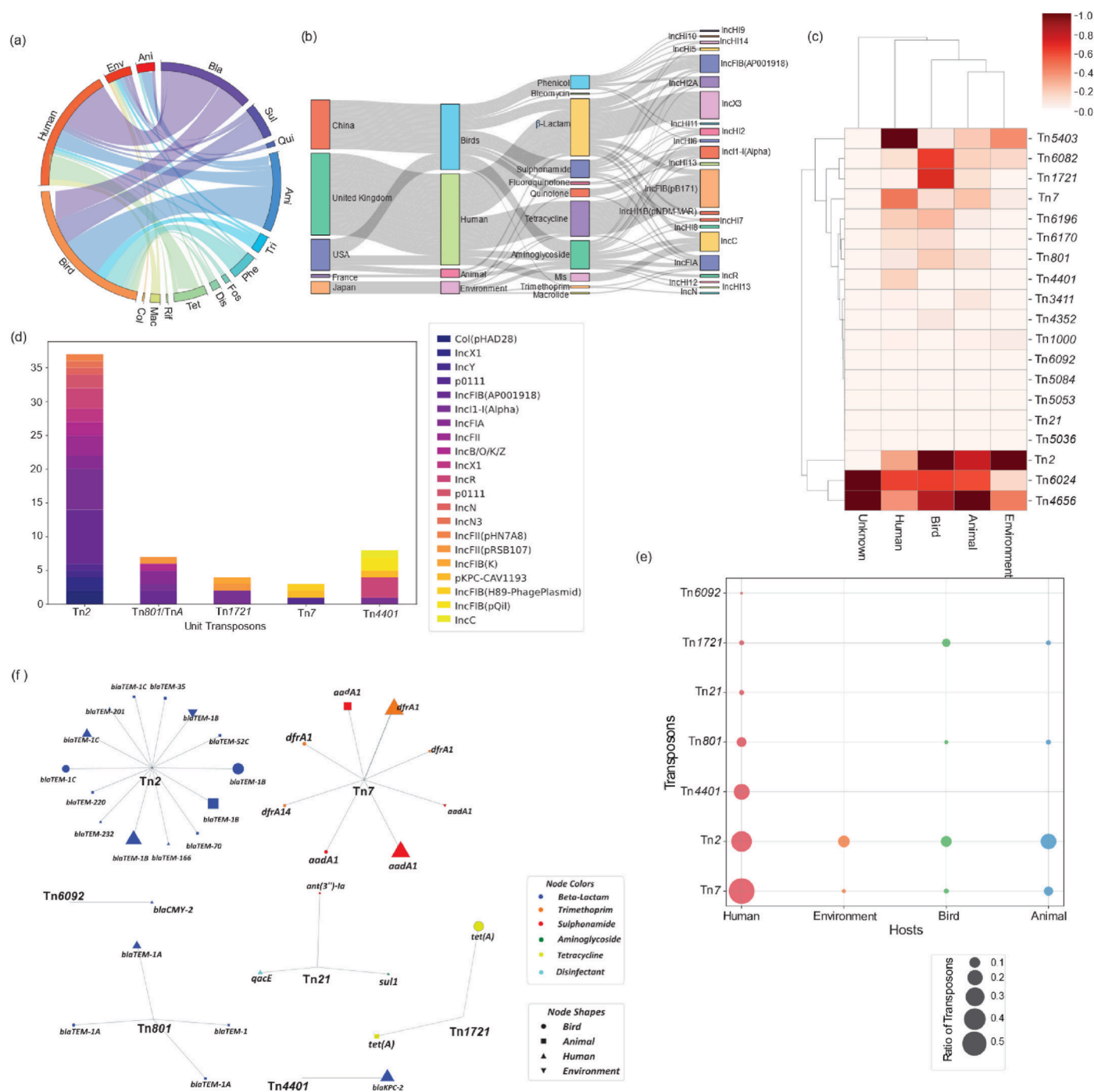


Fig. 5. Abundance of ARGs associated with transposons and IS elements: (a) The Circos diagram illustrates the AMR classes conferred by ARGs associated with IS elements across all hosts. The connecting links between segments symbolize the frequencies of ARGs observed in each host type. The abbreviations mentioned are as follows: trimethoprim, Tri; tetracycline, Tet; sulfonamide, Sul; rifampicin, Rif; quinolone, Qui; phenicol, Phe; macrolide, Mac; fosfomycin, Fos; colistin, Col; beta-lactams, Bla; aminoglycoside, Ami; disinfectant, Dis. (b) Sankey diagram representing the association between country origin, host type, IS-associated ARGs clustered into AMR classes and plasmid Inc-type. The connecting links between segments represents the frequencies of ARGs observed in each category. (c) Heatmap depicting frequency of transposons grouped by hosts in columns and transposon type in rows. Two-way clustering was applied to both the columns and rows for organizing and visualizing the data. (d) Stacked bar plot showing the abundance of different types of transposons carried by diverse set of plasmids. (e) Bubble chart showing the relative abundance of different transposons across all hosts. The size of the bubble depicts the abundance of transposons. (f) Network plot showing the association of ARGs with different transposons. The different shapes represent different hosts. The size of each shape depicts the relative abundance of ARGs associated with transposons. The edge connected by shapes is irrespective of any parameter.

diverse association patterns in bird followed by human host (Supplementary Table S11).

A total of 30 IS elements were identified in association with ARGs, with two major clusters linked to most of the AMR classes. The IS26 served as hub of diverse set of ARGs conferring maximum resistance to

aminoglycoside and β -lactam in all the hosts (Figs. 6 and 7). The ISVsa3 (194, 0.78%), although less abundant, was identified as the hub of ARGs resistant to aminoglycosides, sulfonamides, and phenicols, primarily found in bird hosts. (Fig. 6). Some IS elements tend to be biased towards specific source (e.g., ISAba1 and IS3 in bird isolates), while others (e.g.,

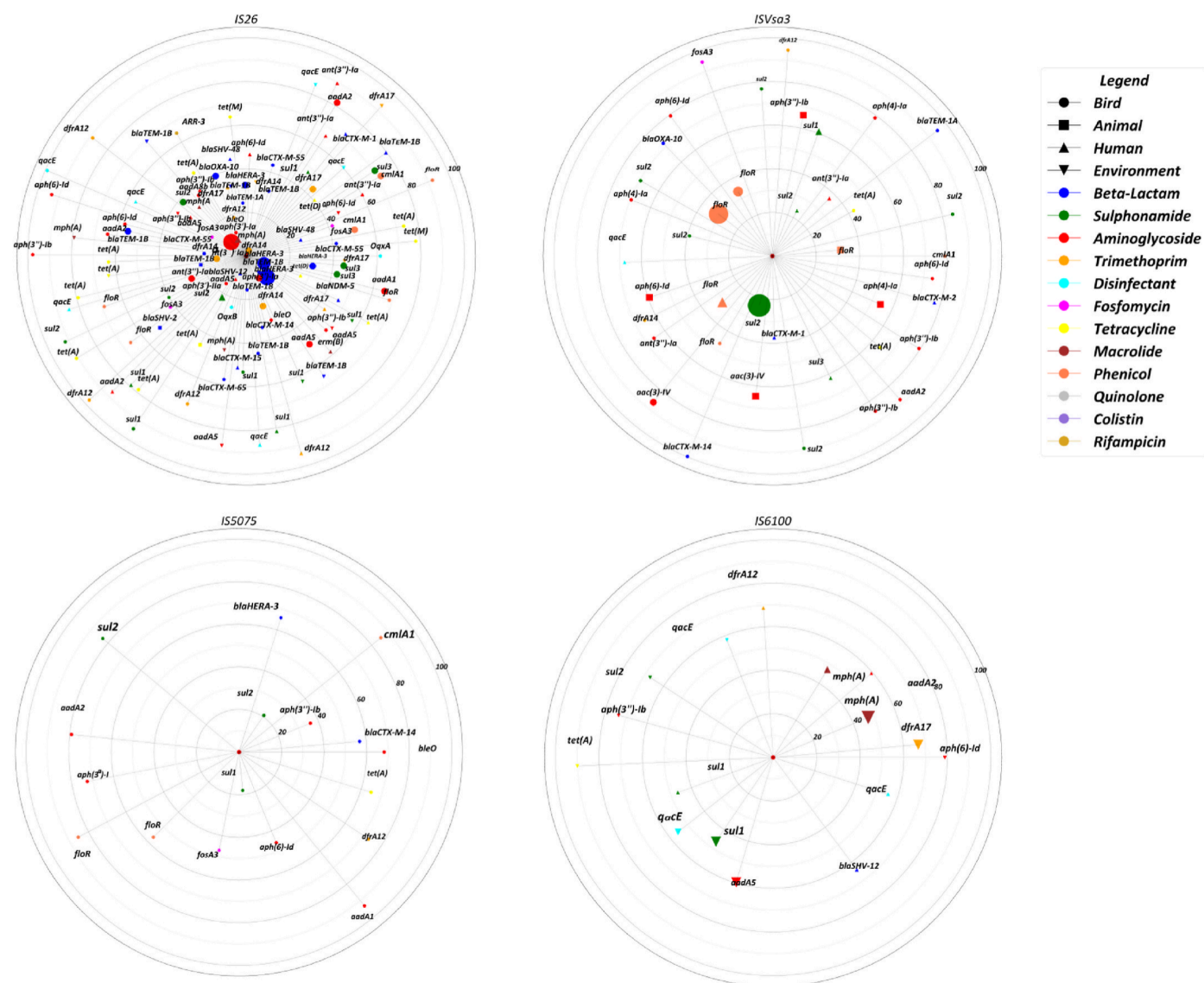


Fig. 6. Network plot of each IS element and their associated ARGs across all hosts. The different shapes represent different hosts. The size of each shape depicts the relative abundance of ARGs associated with IS element. The edge connected by shapes represents the distance between ARG and IS element mentioned besides each rim. The distance is represented in square root transformation with a maximum distance of 10 kb.

IS26 and ISVsa3 were widespread (Supplementary Fig. S5). The IS629 was abundant but had low association with ARGs. The ISEc59, ISKpn24, IS1133, and ISAeme22 predominantly carried ARGs related to aminoglycoside resistance. The IS5, ISEc9, and ISKpn14 were primarily associated with ARGs linked to β -lactam antibiotics.

A total of 137/464 combinations were in the category of ‘very strong association’ linked with 15 IS elements, majorly IS26 with almost 1/3rd covered by genes of β -lactam mainly *bla*_{NDM-5} (23/64), *bla*_{HERA-3} (11/64), *bla*_{TEM-1} (7/64). A total of 27 *bla*_{NDM-5} was found associated with IS elements, of which 23 were within 1000 bp majorly linked with IS5 and ISAbi125 and 17 were carried by IncX3 plasmid found majorly in UK humans followed by China birds (Supplementary Fig. S5, Figs. 6 and 7 & Supplementary Table S11). Almost 50% of IS-ARG combinations (69/137) were found at a distance within 200 bp, such as *bla*_{CTX-M-15}/*bla*_{CTX-M-14}/*bla*_{CTX-M-1}, ISEc9, *bla*_{NDM-5}, IS5, *aph*(3)-Ia, IS26 suggesting a higher likelihood of their transmission as a cassette between different hosts. The smallest distance accounts to 1 bp between *ant*(3)-Ia of aminoglycoside class associated with IS26 and 3 bp distance between *bla*_{CTX-M-65} and IS102 but had occurred only once and hence, could be a chance factor. Therefore, to validate, BLASTn was done, showing many

incidences with 100% match with the sequence present in *Enterobacteriaceae*, indicating the possibility of HGT beyond genus level. A total of 121/464 and 99/464 combinations were in the category of ‘strong association’ and ‘moderate association’ respectively.

The genetic distance around the most prevalent ARGs associated with IS elements namely *aph* genes, *sul2*, *bla*_{NDM-5} and *bla*_{CTX-M} variants was explored. These ARGs were situated at varying distances, spanning from 20bp to 9 kb from the IS elements except *bla*_{NDM-5} and *bla*_{CTX-M} variants, which displayed a consistent pattern, mostly residing within a range of 2 kb and 5 kb from the IS elements respectively. In most cases, both *bla*_{NDM-5} and *bla*_{NDM-7} were consistently identified at an equidistant position from the IS elements, particularly ISAbi125 and IS5.

The mobility of IS-associated ARGs was assessed by investigating the combinations of ARG and IS element based on consensus distance found across diverse hosts and countries (Fig. 8 & Supplementary Fig. S6). Mostly, the combinations with consistent distance found in different hosts were in the category of ‘very strong association’ and ‘strong association’. There was a total of 16 such IS-ARG combinations with consensus distance found in different hosts and countries, with 14/16 combinations within 3 kb distance. The IS5_ *bla*_{NDM-5} combination was

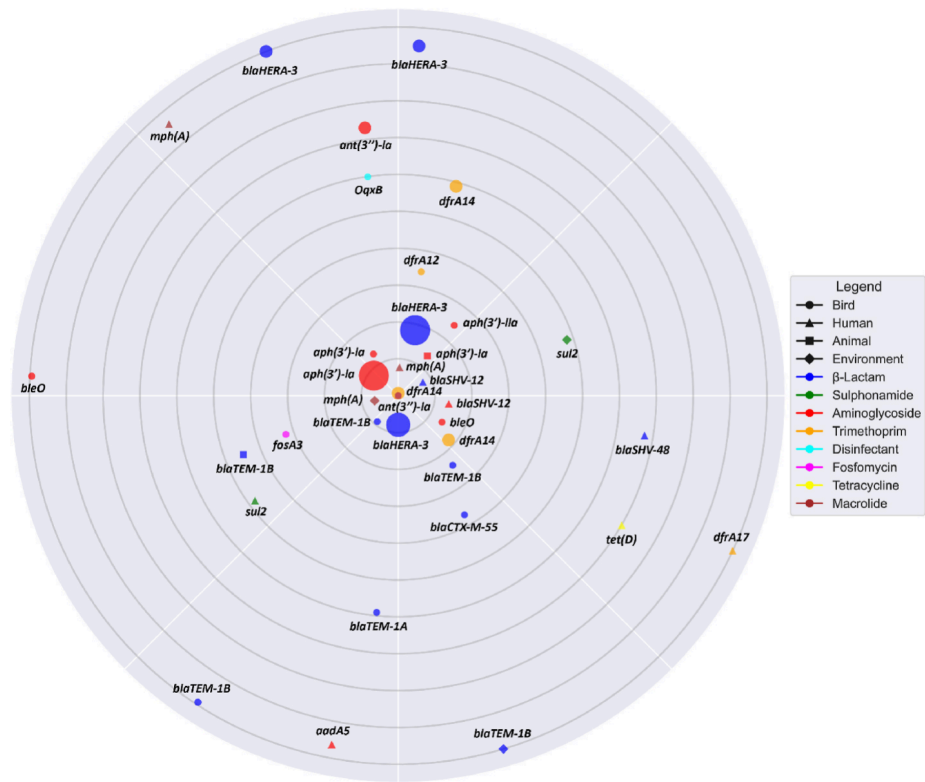


Fig. 7. Radial plot of IS26 element and its associated ARGs across hosts. The different shapes represent different hosts. The size of each shape depicts the relative abundance of ARGs associated with IS element. The edge connected by shapes represents the distance between ARG and IS element, with maximum distance of 1 kb. Each circumcentre represents a gap of 100 bases.

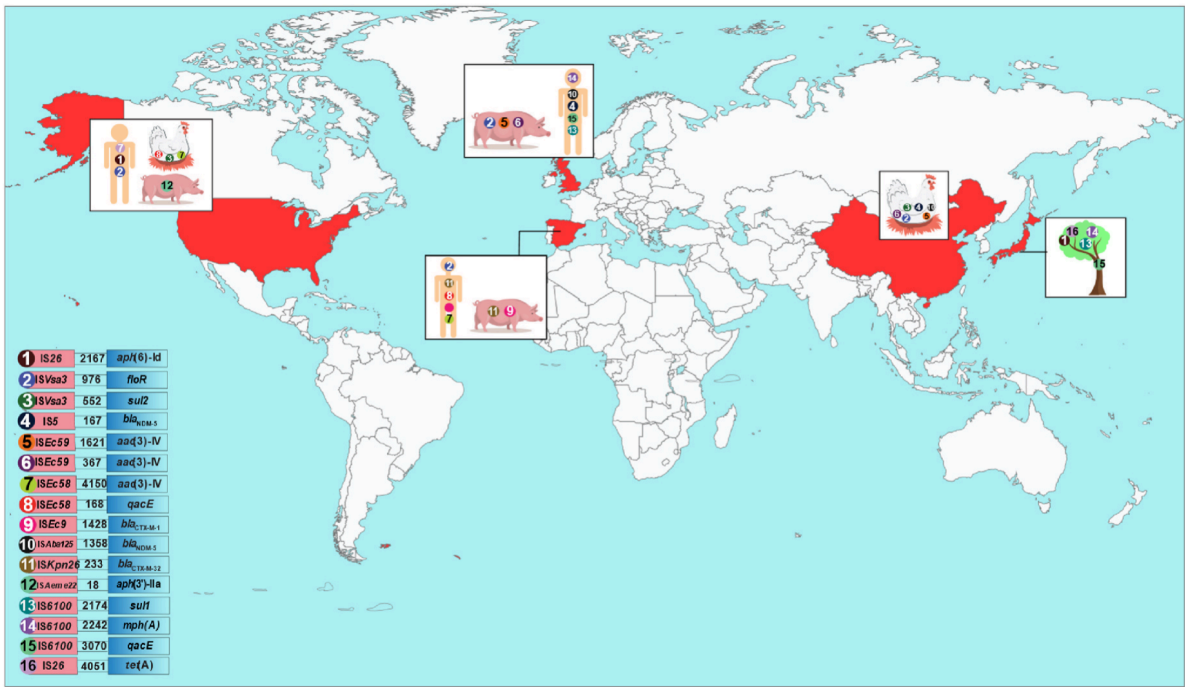


Fig. 8. World map depicting the possible transmission of IS-ARG combinations in different countries and hosts. Same numbers represent the presence of same IS-ARG combinations.

found in UK human and in environmental samples isolated from Japan. The *floR* associated with ISVsa3, was detected in USA, UK and China isolated from human, animal and bird hosts respectively (Fig. 8).

4. Discussion

Gram-negative bacteria including *E. coli* acts as a crucial reservoir of ARGs and MGEs, facilitating the acquisition and dissemination of

resistance. (Xu et al., 2022a,b). Plasmids, the autonomously replicating extrachromosomal element, promote horizontal gene transfer of ARGs, enhancing the transmission of ARG worldwide (Qin et al., 2024). The plasmids, MGE and host origin are important interlinked factors in the exploration of the global spread of ARGs. Their association significantly influences transmission likelihood of ARGs across diverse hosts and regions (Evans et al., 2020). This study focuses on assessing ARG prevalence and their linkage with MGEs using publicly available *E. coli* whole genome sequences (n = 2199). Analysis of the physical proximity between ARGs and MGEs aims to predict their transfer patterns across different hosts and countries.

Human, animal, bird and environment are pivotal in disseminating antibiotic resistance through the transfer of resistant bacteria across different environment, posing a global challenge. (Brooks et al., 2023; Koga et al., 2019; Mohapatra et al., 2023; Murray et al., 2021). Though bird samples were only a quarter of the human samples, birds were identified as a significant reservoir of IS-associated ARGs. Poultry production is a significant industry worldwide, heavily reliant on antimicrobials in feed, regardless of infection (Grzinić et al., 2023). This practice elevates the selective multiplication of AMR *E. coli*, particularly in dense poultry population (Cao et al., 2022). Driven by a strong demand for poultry products, global poultry numbers exceed 26 billion, making up approximately 70% of all avian biomass (Mageiros et al., 2021). Wild birds, particularly migratory species, can travel long distances, potentially carrying ARGs from contaminated environments, agricultural sources, or human habitats, often crossing borders. (Liu et al., 2024). These birds can act as ARG reservoirs, spreading them across diverse geographical regions and introducing them into new microbial communities (Yuan et al., 2021). They act as conduits for ARG transmission between wildlife, human-associated bacteria and environmental bacterial populations (Liu et al., 2024; Russo et al., 2022; Yuan et al., 2021). This raises concerns about the potential for “spill-over” of resistant *E. coli* from bird hosts to human and other sources, thus complicating the global challenge of resistance against antibiotics (Aworh et al., 2021; Baker et al., 2024).

β -lactams and aminoglycosides are two commonly used classes of antibiotics for treating bacterial infections (El-Far et al., 2021; Lima et al., 2020). However, their widespread use has led to significant resistance among bacteria, complicating treatment efforts (El-Far et al., 2021; Medugu et al., 2023). This study performed clustering analysis of ARGs into their corresponding AMR classes, revealing that aminoglycoside and β -lactam antibiotics exhibit the highest levels of resistance. Rubin et al. studied the ARG abundance in 102 human samples (Rubin et al., 2020). More than 70% of the samples contained ARGs corresponding resistance to β -lactam and aminoglycoside (Rubin et al., 2020). Another study showed the similar results in the environment samples (Obayiwana et al., 2021). ARGs conferring resistance to β -lactam antibiotics, such as *bla*_{CTX-M} (*bla*_{CTX-M-15-14}) and *bla*_{NDM} (*bla*_{NDM-5}, *bla*_{NDM-1}) is particularly worrisome due to their rapid spread, creating an alarming situation (Huang et al., 2024; Li et al., 2024a,b; Rana et al., 2022; Yu et al., 2024). The prevalence of *bla*_{NDM-5} has been steadily increasing, contrasting with a decline in *bla*_{NDM-1}, consistent with our findings (Li et al., 2024a,b).

While most ARGs might be present with limited transfer potential to bacteria, many have enhanced their mobility by associating with small transposable elements like IS elements (Berglund et al., 2023; Partridge et al., 2018). Once mobilized, ARGs can spread rapidly, especially under selection pressure from antibiotics (Emara et al., 2023; Yu et al., 2024). Two major IS clusters found associated with ARGs were identified, IS26 and ISVsa3, with many exhibiting strong associations (Acman et al., 2022; Harmer and Hall, 2024; Jiang et al., 2024; Lewis et al., 2023; Tang et al., 2024; Yu et al., 2024). The IS26 (820 bp) is highly effective in spreading ARGs in Gram-negative bacteria by recruiting ARGs or ARG clusters into mobile gene pools (Harmer et al., 2014; Tang et al., 2024). The IS26 can enhance the expression of nearby resistance genes by providing a strong promoter (Aihara et al., 2024; Harmer et al., 2014). It

facilitates their movement to new locations by forming pseudo-compound transposons or translocatable units, which can then be mobilized further (Harmer and Hall, 2021). IS26 displays a preference for random insertion and thus has more opportunities for insertion of ARGs into the genome (Tang et al., 2024). The ISVsa3, an IS91-like MGE, mobilizes through a process called rolling-circle replication (Del Pilar Garcillán-Barcia et al., 2001; Marazzato et al., 2022; Toleman and Walsh, 2010). This mechanism enables the co-mobilization of additional sequences, such as ARGs, positioned upstream of their transposase genes (Lewis et al., 2023). Due to this unique mode of transposition, ISCR elements can efficiently mobilize a wide range of AMR gene classes (Lewis et al., 2023). These IS elements likely retain and accumulate ARGs, promoting co-selection of bacteria under antibiotic pressure and enabling them to acquire more ARGs (Harmer and Hall, 2024; Jiang et al., 2024; Pong et al., 2019). This might also indicate that there is little to no discernible preference between these IS elements and ARGs (Harmer and Hall, 2020; Leekitcharoenphon et al., 2021; Lewis et al., 2023; Shafiq et al., 2022).

MGEs such as transposons and plasmids play a crucial role in the horizontal transfer of ARGs, facilitating their spread across diverse bacterial populations and environments (Coluzzi et al., 2022; Lu et al., 2024). The IncX3 plasmids are narrow host-range plasmids primarily found in *Enterobacteriaceae*, particularly *E. coli* (Qin et al., 2024). They are known for their efficient conjugation ability, high stability and no fitness costs (Guo et al., 2022). Over the past decade, IncX3 plasmids have been strongly associated with the transmission of *bla*_{NDM} genes, which confer carbapenem resistance (Guo et al., 2022). The IncFIB (pB171) plasmid plays a role in ARG dissemination similar to that of IncX3 (Takayama et al., 2020). Its backbone regions are relatively small compared to its accessory regions, which contain high number of mobile elements (Nazir et al., 2020). These elements act as drivers of horizontal gene transfer, promoting the spread of ARGs such as *tet*(B), across diverse hosts and environments on a global scale (Arnold et al., 2021; Nazir et al., 2020).

The association between Tn2 and IncFIB plasmids further highlights their critical role in spreading ARGs across bacterial populations (Lu et al., 2024). In the present study, Tn2, the most widely distributed transposon, was carried by 15 different plasmids, with a particular affinity for IncFIB plasmids. The IncFIB plasmids, widely observed in *E. coli*, often carry resistance determinants and are highly associated with hospital-acquired infections (Xu et al., 2022a,b). The Tn2 efficiently carry and mobilize *bla*_{TEM} gene variants, contributing to β -lactam antibiotic resistance (Leekitcharoenphon et al., 2021; Malesa et al., 2024). This significant relationship between Tn2 and IncFIB plasmids emphasizes the need for monitoring these genetic elements, as they contribute to the robustness and spread of resistance traits in bacterial population across human, animal and environment (Li et al., 2023; Malesa et al., 2024).

Bacteria survive in different hosts by acquiring resistance genes via HGT, typically present on MGEs like IS elements, giving adaptive phenotypic flexibility (Arnold et al., 2021; Consuegra et al., 2021). Our study highlights the specific preferences of certain mobile elements for acquiring and transferring AMR genes. Certain IS elements exhibited a tendency to be situated near specific ARGs (Zhao et al., 2020). The IS5 primarily harboured *bla*_{NDM} genes (Acman et al., 2022). However, previous studies have also reported the presence of other β -lactam genes carried by IS5 (Kurpiel and Hanson, 2011; Tian et al., 2020). The ISKpn14 carried ARGs (*rmtC*, *bla*_{NDM-4}, *bla*_{NDM-1}) conferring resistance aminoglycoside and β -lactam (Partridge et al., 2018). There were IS elements that had fewer associations with ARGs (Leekitcharoenphon et al., 2021). The ISEc1 and IS629 (member of IS3 family), have shown wide prevalence found across all host types within *E. coli* genome (Founou et al., 2022; Leekitcharoenphon et al., 2021). However, they showed no ARG associations, possibly because some IS elements serve various other important roles in prokaryotic genomes beyond ARG mobilization (Khedkar et al., 2022; Kusumoto et al., 2011; Schrader and

Schmitz, 2019; Weisberg and Chang, 2023). These elements contribute to genetic diversity and adaptation by facilitating insertions, deletions, and duplications, which drive genome evolution (Siguier et al., 2014). By integrating near regulatory regions, IS elements can also modulate gene expression, impacting bacterial physiology and stress responses (Consuegra et al., 2021). They form pathogenicity islands, allowing bacteria to acquire traits like virulence genes, metabolic capabilities, environmental resilience, which can spread across hosts and environments (Weisberg and Chang, 2023).

Examining the context of IS elements surrounding ARGs is vital for understanding the molecular mechanisms of ARG mobilization in diverse bacteria (Razavi et al., 2020). We consider ARGs associated with MGEs as ‘cassettes’ and utilize comparative genomics methods to investigate their propagation patterns based on their genetic framework. Recently, the association of IS elements and ARGs have gained much attention (Cao et al., 2022; Evans et al., 2020; Leekitcharoenphon et al., 2021; Nielsen et al., 2022; Razavi et al., 2020). We hypothesized that closer IS-ARG proximity, enhancing the likelihood of co-mobilization, reflects a stable and long-term association (Nielsen et al., 2022). Thus, in present study, a comprehensive analysis of the physical distance between the ARGs and IS elements was performed. Network analysis graph revealed strong associations between specific IS-ARG combinations. Most β -lactam and aminoglycoside genes are located within 200bp of IS elements. Of concerns, are combinations like *bla*_{CTX-M-1/14/15}*I*Sec9 and *bla*_{NDM-5}*IS*5, which facilitate the rapid spread of AMR genes across different bacteria (Decano et al., 2019; Rana et al., 2022; Wailan et al., 2016; Yu et al., 2024). Recently, *I*Sec9 was implicated in *bla*_{CTX-M} transposition, emphasizing their close proximity and enhanced transmission potential (Medugu et al., 2023).

The threshold (<10 kb) is significant as previous studies suggest that the proximity between ARG and MGE within this range increases the likelihood of these clusters to move as a unit during HGT, thereby enhancing the transmission potential of ARGs (Cao et al., 2022; Toth et al., 2021; Trosvik et al., 2024). This results in increasing ARG dissemination across diverse bacterial species in human, animal and environmental globally (Boolchandani et al., 2022; Cao et al., 2022; Evans et al., 2020; Karkman et al., 2020; Partridge et al., 2018; Razavi et al., 2020). In various instances, ARGs were found at variable distance with IS elements, ranging from 1 kb to 10kb suggesting the capability of the IS elements in mobilizing the ARGs is irrespective of the physical distance (Leekitcharoenphon et al., 2021; Razavi et al., 2020). The similar distance of *bla*_{NDM-5} and *bla*_{NDM-7} from the IS elements might indicate a selective advantage for these specific genetic arrangements (Huang et al., 2024; Qin et al., 2024; Yang et al., 2022). These genetic structures may be more stable or have higher transmission potential, allowing them to persist in *E. coli* population (Wailan et al., 2016). Analysing distances in the 1bp-10kb range allows a detailed understanding of which ARGs are frequently positioned near specific IS elements. This might reveal the potential preferences of IS elements for transferring particular ARGs (Bailey et al., 2011; Khedkar et al., 2022; Partridge et al., 2018). This distance-based analysis helps determine the likelihood of an ARG being located within a specific range of proximity to IS elements (Khedkar et al., 2022; Partridge et al., 2018).

The information regarding the presence of IS-ARG combinations across different countries and hosts has not been reported in the previous studies (Leekitcharoenphon et al., 2021; Nielsen et al., 2022). Our research provides a comprehensive analysis of the presence of the same combination of ARG and IS element with consensus distance, found in different countries and hosts such as *IS5_bla*_{NDM-5}, *ISVsa3_floR*. This indicates that some AMR genes tend to be located at a consensus distance from the IS elements in *E. coli* genome (Partridge et al., 2018). Studies support these associations, showing that *bla*_{NDM-5} is globally distributed with conserved genetic background and is mobilized by *IS5* (Yang et al., 2022). The *IS5_bla*_{NDM-5} often resides on highly conjugative IncX3 plasmids, facilitating its spread effectively across bacterial populations (Li et al., 2023; Yang et al., 2022). Similarly, *ISVsa3* is strongly linked

with *floR* genes in isolates from diverse regions, across various sources (Ayyappan et al., 2024; Lewis et al., 2023; Zhou et al., 2024). These strongly associated genetic combinations, within 1 kb range, are compact and stable structures (Khedkar et al., 2022; Lewis et al., 2023; Partridge et al., 2018). The proximity of IS-ARG clusters results in shorter DNA fragments, facilitating efficient HGT through conjugation, especially under antibiotic selection pressure (Cao et al., 2022; Khedkar et al., 2022; Sheppard et al., 2016). Global transmission pathways such as international travel, trade and agriculture further support the dissemination of these resistance elements across various bacterial communities and environmental niches (Tang et al., 2024).

Mostly, these IS-ARG combinations were within 3 kb separation (87.5%, 14/16). This suggests that lesser the distance between the IS elements and the ARGs, more are the chances of their transmission across different hosts. Hence, the compact arrangement of ARGs and the IS elements is crucial for successful dissemination. This could also indicate potential for HGT between different genetic elements, such as from a plasmid to the chromosome or vice versa, or between different bacterial strains.

5. Conclusion

The study highlights the complex landscape of ARGs and MGEs prevalence in *E. coli* across diverse hosts and geographical locations. It underscores the critical role of MGE proximity to ARGs in influencing their transmission potential. Our analysis reveals significant patterns in the distribution of ARGs, with human samples showing the highest diversity and abundance. However, bird hosts harboured the maximum IS-associated ARGs. The *IS26* and *ISVsa3* elements carried a diverse range of ARGs, primarily linked to aminoglycoside and β -lactam resistance, highlighting the persistence and spread of resistance. The close association of these ARGs with insertion sequences, suggests a strong potential for horizontal gene transfer, particularly in human and bird populations. Notably, the identification of plasmids and IS elements as key players in the dissemination of resistance genes further emphasizes the role of MGEs in the global spread of AMR. Our study highlights a potential approach to combat AMR spread by focusing on the genetic context of mobile ARGs. Identifying high-risk ARGs linked with MGEs can target AMR hotspots for management and development of GIS-based surveillance for global AMR control.

CRedit authorship contribution statement

Chanchal Rana: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vaibhav Vikas:** Visualization, Validation, Software, Data curation. **Saraswati Awasthi:** Software, Methodology. **Devika Gautam:** Writing – review & editing, Validation, Methodology. **Ashutosh Vats:** Validation, Software. **Shi-veeli Rajput:** Validation, Methodology. **Manisha Behera:** Validation. **Ashutosh Ludri:** Resources. **Anupam Berwal:** Resources. **Dheer Singh:** Funding acquisition. **Sachinandan De:** Writing – review & editing, Validation, Supervision, Methodology, Investigation, Conceptualization.

Ethical approval

Not Required.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chemosphere.2024.143808>.

Data availability

All python codes used in the study have been deposited online at <https://github.com/vaibhavvikas/ecoli-genome-analysis>

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